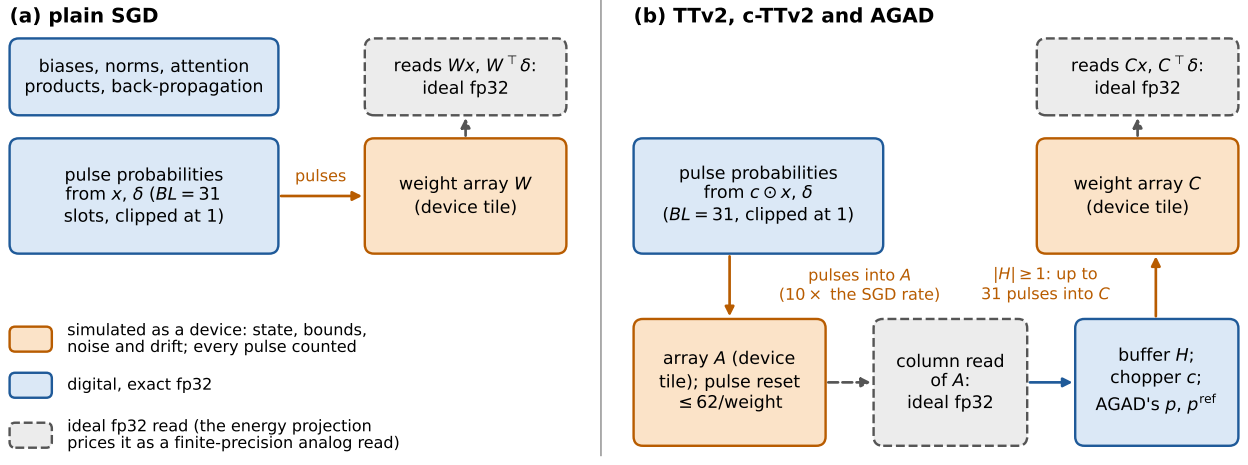
(a) plain SGD
biases, norms, attention products, back-propagation
reads Wx, $W^\top\delta$: ideal fp32
pulse probabilities from x, δ ($BL = 31$ slots, clipped at 1)
pulses
weight array W (device tile)
simulated as a device: state, bounds, noise and drift; every pulse counted
digital, exact fp32
ideal fp32 read (the energy projection prices it as a finite-precision analog read)
(b) TTv2, c-TTv2 and AGAD
reads Cx, $C^\top\delta$: ideal fp32
pulse probabilities from $c \odot x$, δ ($BL = 31$, clipped at 1)
weight array C (device tile)
pulses into A (10 × the SGD rate)
$|H| \geq 1$: up to 31 pulses into C
array A (device tile); pulse reset ≤ 62/weight
column read of A: ideal fp32
buffer H; chopper c; AGAD's p, p^{ref}